

ARAV GUPTA

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Enthusiastic engineering student with practical experience in IoT, embedded systems, robotics, and embodied agents, offering expertise in ROS2, computer vision, multi-agent task allocation, sensor integration, and real-time data acquisition. Experience working with multi-agent Vision-Language Navigation systems and Collaborative Embodied Agent problems. Thorough knowledge in reinforcement learning algorithms, routing problems and transformer architectures. Proven capabilities in the design and deployment of field-ready systems, including LoRa mesh networks and precision agriculture sensors. Seeking various opportunities to utilize technical and problem-solving skills in research and development.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering Birla Institute of Technology and Science, Pilani Current CGPA: 9.7/10.00	August 2023 - ongoing
CBSE, Class XII R.N. Podar School, Khar, Mumbai, 95.2%	June 2021 - April 2023
ICSE, Class X Hiranandani Foundation School, Powai, Mumbai, 98%	June 2013 - April 2021

SKILLS AND INTERESTS

Interests	Robotics, Automobile, Computer Vision, Internet of Things, Embedded Systems Design, Embedded Systems Software, Drone Technology, Vision-Language Navigation, Reinforcement Learning, Transformer architectures, Embodied Agents, Robot Simulation
Technical Skills	ROS2, Gazebo , RViz, PX4 Simulation, Python (numpy, tensorflow, torch, clip), C++, Keil, Arduino , ESP32, RaspberryPi, OpenCV, YOLO, MATLAB, Nav2 Stack

PROJECTS

Internet of Things-based Soil Sensing Module <i>Project as an intern at RRSC-N, ISRO</i>	May 2025 - July 2025
· A project designed for application in Precision Agriculture · Low-cost, modular IoT module for sensing soil moisture and nutrients, with backhaul using LoRaWAN and 4G. · Optimization in terms of power consumption, localization and server integration, while maintaining a low cost factor	
Internet of Things-based Air Quality Index measuring module <i>Project as an intern at RRSC-N, ISRO</i>	May 2025 - July 2025
· Made a low-cost, portable AQI measuring module running off CPCB model for AQI measurement · Configured all sensors after intensive tests, and provided with 4G backhaul to server. · Conducted field tests by travelling to different parts of city in order to calibrate sensors with accurate values.	

Object tracking swarm drone system in Gazebo

August 2025- October 2025

Part of project under faculty at BITS Pilani

- Created a custom, robust object tracking model using OpenCV
- Simulated tracking using PX4 drones in Gazebo using ROS2, MicroXRCE-DDS, MAVROS.
- Scaled to support multiple drones tracking individual objects, with anti-collision algorithm
- Plans to implement trust algorithms to system for practical applications.

Bandwidth-Constrained Cooperative Vision-Language Navigation

May 2026

Independent Project

- Sole author of a research paper currently under review
- Proposed hindsight communication gating, a BCE-supervised communication policy that eliminates high-variance REINFORCE training for multi-agent communication
- Introduced bandwidth-constrained cooperative VLN and demonstrated a synchronisation-driven communication regime, where agents communicate early to align hidden representations rather than recover from uncertainty
- Designed an asymmetric paired-agent dataset and conducted hidden-state alignment analysis showing 260% higher alignment efficiency than random communication at matched budgets

Cooperative Embodied Language Agents

Ongoing

Part of project under faculty at BITS Pilani

- Developing a belief-aware multi-agent embodied AI framework for collaborative household task completion under partial observability
- Designed probabilistic belief graphs, confidence-aware planning, and adaptive communication strategies for cooperative embodied agents
- Evaluated greedy and LLM-guided planning policies on VirtualHome/CWAH benchmarks using success rate, SPL, communication efficiency, and coordination metrics

Autonomous Mobile Robot for Industrial Applications

Ongoing

Collaborative effort with ASME (American Society of Mechanical Engineers) chapter at BITS Pilani, for a robotics competition

- Design of a autonomous mobile robot, able to navigate dynamic obstacles and perform pick and place operations, enabling efficient sorting and inventory management
- Helped ASME (BITS Pilani) design a system for reading fiducial markers using OpenCV via a ESP32-S3 Camera Module, communicating data to a RaspberryPi via MQTT
- Setting up system for employing SLAM (Simultaneous Localization and Mapping) techniques using 3D-LiDAR, along with Nav2 stack of ROS2 for motion.

Deep Reinforcement Learning based Rover Routing

June 2026

Proposed solution for hackathon

- Designing a Deep Reinforcement Learning algorithm for routing a battery constrained rover to explore potential ice regions in lunar permanently shadowed regions (PSR)
- Utilized attention model for policy generation, REINFORCE gradient for training, in order to learn complex relations between candidate nodes.
- Worked with a team of 3 other talented students to develop pipeline, working on improving pipeline for future use.

RESEARCH PUBLICATION

Early to Share, Late to Save: Synchronisation-Driven Communication Gating in Bandwidth-Constrained Cooperative VLN

Arav Gupta.

Accepted at the IJCAI 2026 GLOW Workshop (to appear in Springer CCIS), 2026.

INTERNSHIP/TRAININGS/CERTIFICATES

Engineering Intern at Regional Remote Sensing Centre- North, ISRO May 2025 - July 2025

- Designed two low-cost, modular IoT modules for soil sensing and AQI detection with local storage as well as backhaul.
- Conducted field testing and calibration of both systems for close-to-ideal responses.
- Designed a LoRaWAN based system with 4G backhaul for data transmission and server updation

Research Intern at Indian Institute of Technology - Bombay May 2026 - Current

- Implementing and benchmarking transformer-based routing and combinatorial optimization algorithms
- Investigating improvements to attention mechanisms and reinforcement learning methods for vehicle routing and multi-agent planning problems

Robotics Automation Workshop August 2023

Organized by: *RoboAI*

Workshop covering multiple aspects of robot design pipeline and application, with interdisciplinary knowledge.

ROS2 Advanced Topics - Certificate August 2025

Udemy

Issued for completing course covering advanced topics in ROS2

ROS2 Nav2 and SLAM - Certificate December 2025

Udemy

Issued for completing course detailing usage of Nav2 stack and application of SLAM to robots in ROS2

POSITION OF RESPONSIBILITY

Student-Faculty Council Member- EEE Department August 2024 - Present

Department of Electrical Engineering, BITS Pilani

- One of 3 students selected to represent the batch of 2023 graduating from electrical branches, out of 345 students.
- Act as a link between both faculty and students, collecting grievances and feedback from students, compiling them, and representing them in monthly meetings with faculty to help resolve issues and strive for improvements in academics and research.

BOSM Co-ordinator August 2025 - Present

Informalz, department at BITS Pilani

- Responsible for managing informal events at BITS Open Sports Meet (BOSM).
- Handle all payments to vendors for organizing events along with setting up of events.
- Co-ordinate volunteers and department members to organize events with minimal disturbances.

APOGEE Co-ordinator August 2025-Present

Matrix, the Cinema and Literature Club at BITS Pilani

- A college cinema and literature club founded by like-minded individuals to celebrate the rich culture of art, literature, shows and movies around the world.
- Responsible for conducting events related to cinema and literature during APOGEE, the technical fest at BITS Pilani.
- Organize book fairs, mobilize student members to set up events and ensure smooth operation.

AWARDS AND RECOGNITIONS

- **Scholar's Badge** — Hiranandani Foundation School
Awarded for excellent performance in extra-curricular activities as well as academic spheres throughout duration at the institution

REFERENCES

Mr. Akash Goyal

Scientist 'SF', RRSC- N, NRSC, ISRO

Email : akash_g@nrsc.gov.in

Project Evaluator (Sensor System Review)

LINKS

Github: <https://github.com/AravG13>

Linkedin: www.linkedin.com/in/arav-gupta-3a6a2321b